Dynamic Pricing & Elasticity Optimization

AquaPure Systems — Water Treatment Systems

Measuring how price actually drives demand — and converting that into margin-safe, segment-level pricing decisions.

Econometric demand modeling · customer segmentation · constrained price optimization · executive decision support

This project uses synthetic data and anonymized company/product names. It is inspired by a real pricing analytics use case but does not disclose confidential company data, internal systems, customer information, or proprietary business results.

Executive Summary

-1.01	+10.1%	$231,795	79%
Overall price elasticity	Gross-profit uplift	Incremental gross profit	Model fit (R^2)

- Demand is close to unit-elastic overall (-1.01), but varies widely — from -0.62 (Industrial Treatment) to -1.61 (Residential Filters).
- Strongest pricing power sits with **Industrial Accounts** (-0.46) and the **North** region (-0.73).
- Constrained optimization (±10% price band, 30% margin floor) yields **$231,795** incremental gross profit (**+10.1%**) for a small -0.9% revenue trade-off.
- 55 targeted price increases, 1 promotion and 0 discount-reduction actions identified across 125 product × segment × region cells.

The Business Problem

Finance and Marketing need to know how price changes affect demand — but demand responds to many forces at once, so raw price-vs-volume comparisons mislead:

- **Own price & discount depth** — the levers we control.
- **Customer segment** — homeowners behave nothing like industrial accounts.
- **Region** — purchasing power and competition differ by geography.
- **Seasonality & marketing** — campaigns and weather move volume independently of price.
- **Competitor pricing & macro conditions** — external substitution and inflation.

The risk: without isolating the true price effect, blanket discounting quietly erodes margin while price increases are left on the table where customers would have paid more.

Our Analytical Framework

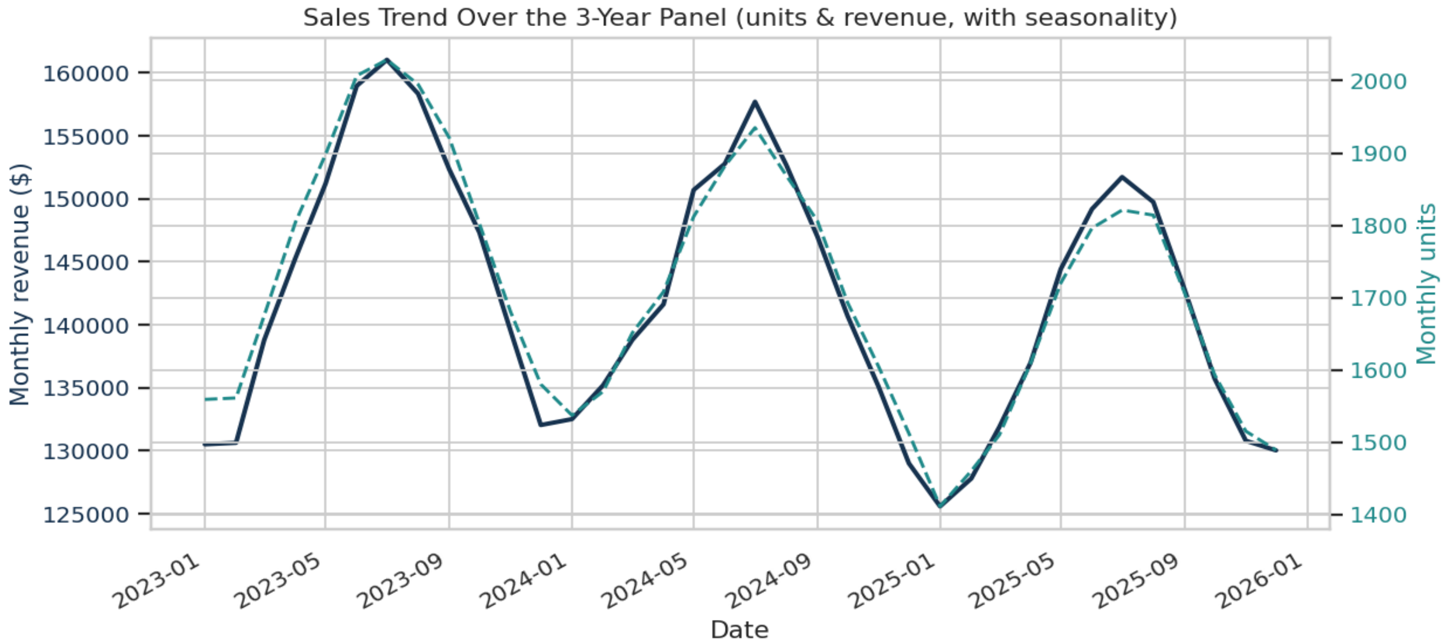
A transparent, econometric pipeline — every step is auditable and reproducible:

1. Data	2. Elasticity	3. Segmentation	4. Optimization	5. Recommendations
Clean panel of sales, price, cost, competitor & macro	Log-log demand regression with fixed effects	K-Means on behavioural pricing features	Constrained search within ±10% & margin floor	Prioritised, margin-safe price actions by cell

We deliberately use **econometrics, not a black box**: coefficients are interpretable, defensible to Finance, and come with confidence intervals.

Data Foundation

A **26,987-row** monthly panel spanning 3 years across 5 regions, 5 customer segments, 5 product categories and 30 products. Each row carries price, discount, cost, margin, units, marketing, competitor and macro indicators.



Monthly units and revenue across the panel — note the recurring seasonality the model must control for.

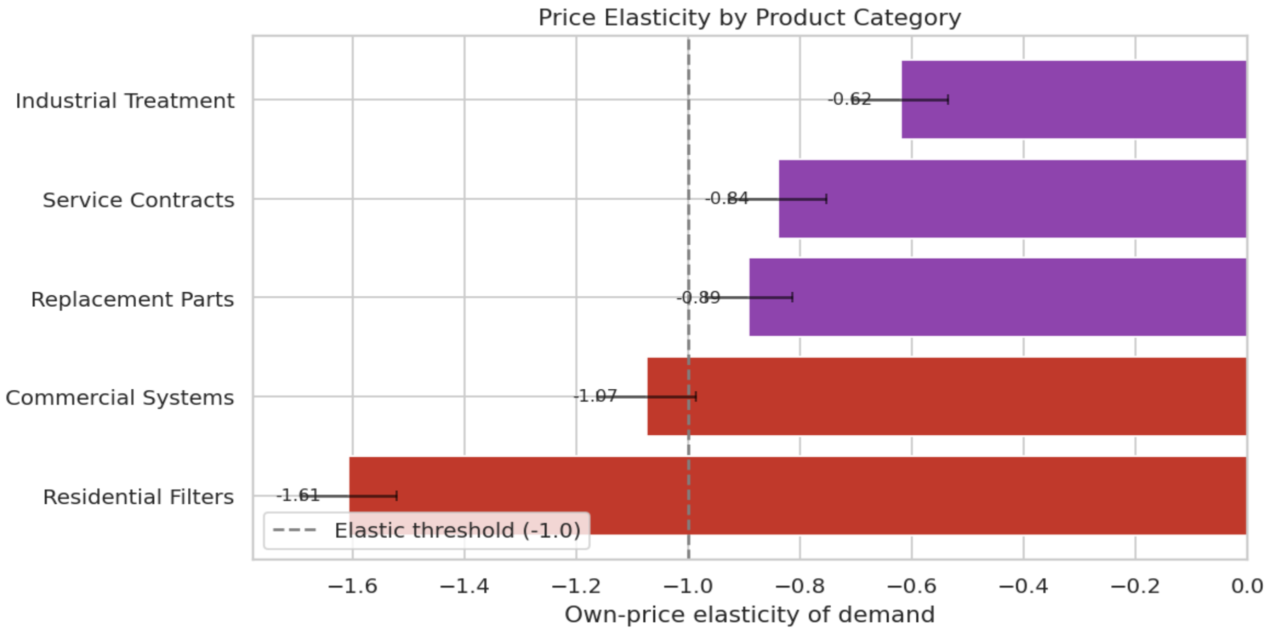
Methodology — Log-Log Demand Model

We estimate a constant-elasticity demand curve. The coefficient on log(net price) **is** the own-price elasticity of demand:

```
log(units) = β■ + β■·log(net_price) + β■·discount + β■·marketing + β■·log(competitor_index) +
macro + region + category + product + segment + month fixed effects + ε
```

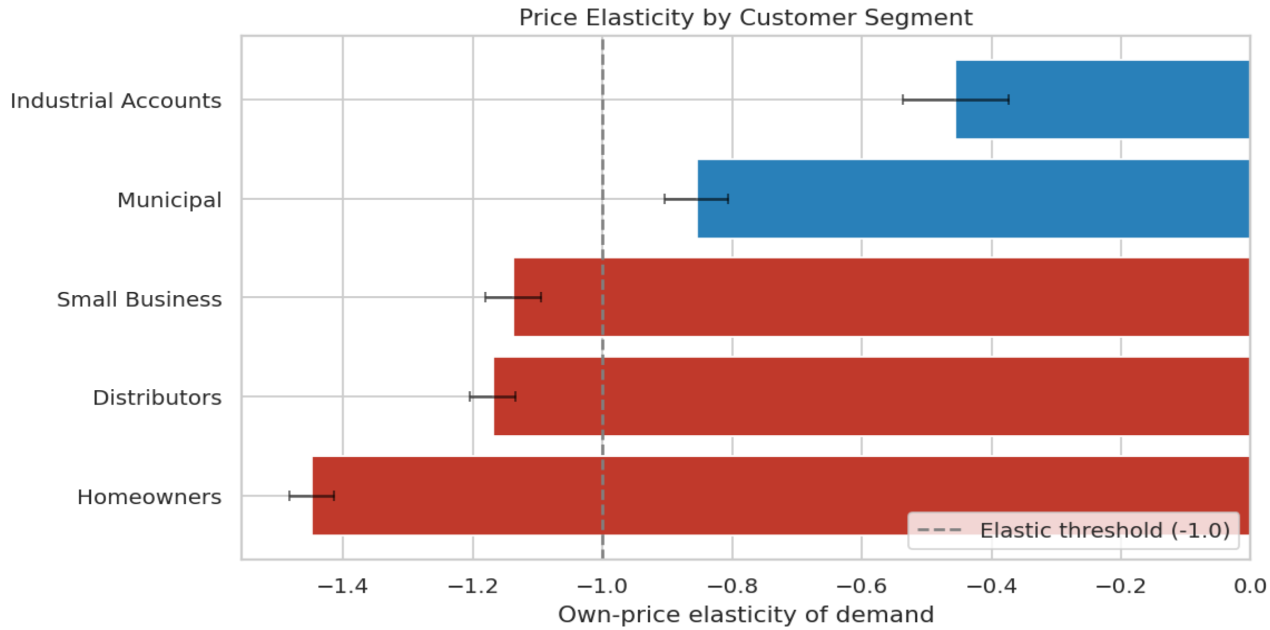
- **β■ = -1.01** overall — a 1% price rise moves demand about 1.01% the other way.
- **Fixed effects** for region, category, product, segment and month strip out confounding from seasonality, purchasing power, product mix and campaign timing.
- **HC3 robust standard errors** guard against heteroskedasticity; all key estimates are statistically significant ($p < 0.01$).
- **Identification:** exogenous list-price variation lets us separate the price effect from the discount lever — $R^2 = 0.80$.

Results — Elasticity by Product Category



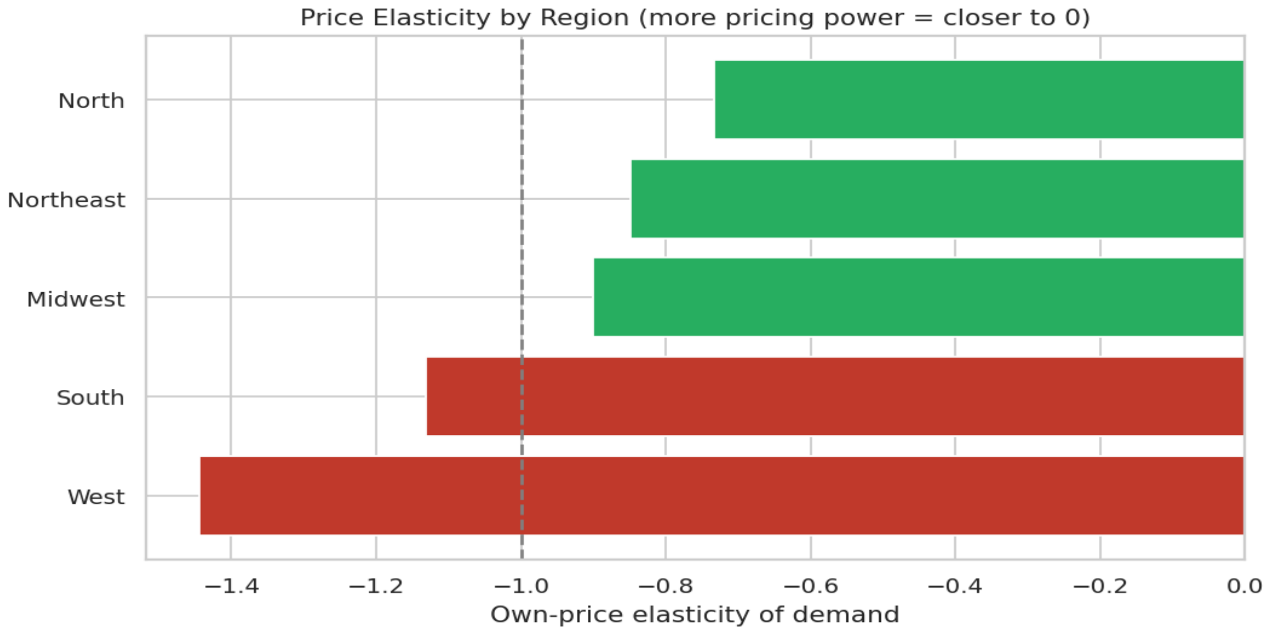
Residential Filters is the most price-sensitive (-1.61); **Industrial Treatment** has clear pricing power (-0.62). Bars left of the dashed line are elastic (volume falls faster than price rises).

Results — Elasticity by Customer Segment



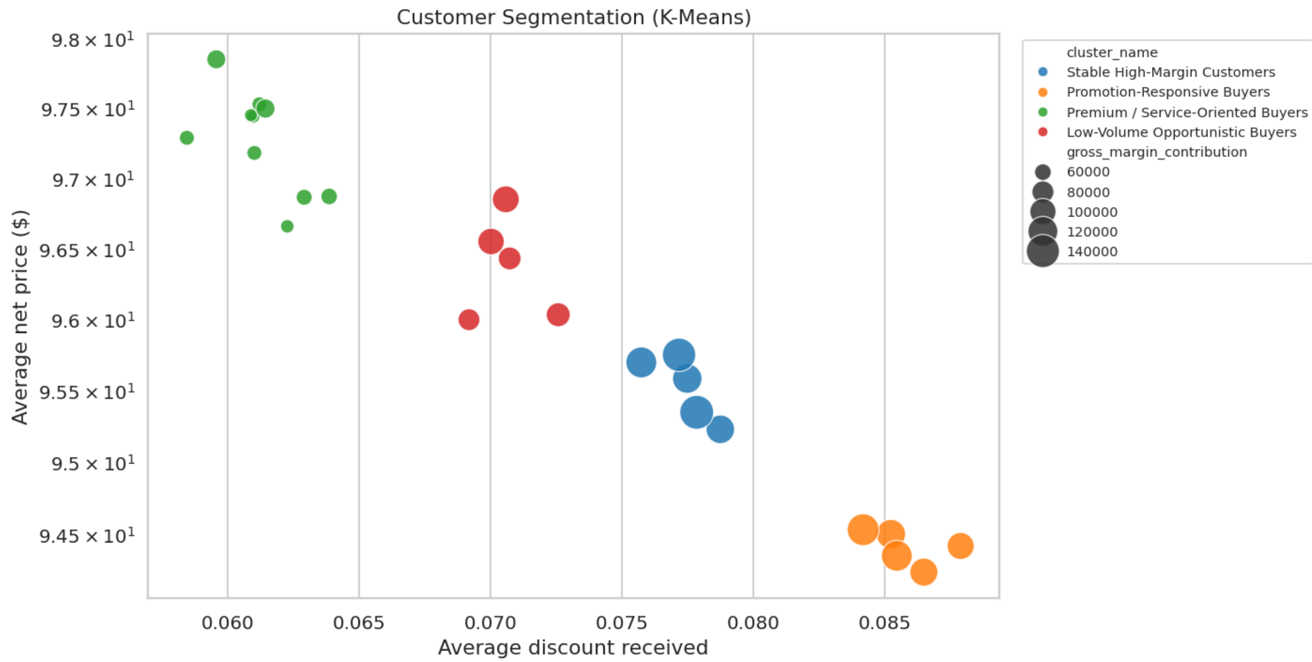
Homeowners are most elastic (-1.45) — protect volume with targeted offers. **Industrial Accounts** tolerate price (-0.46) — a margin opportunity. Whiskers show 95% CIs.

Results — Elasticity by Region



The **North** region supports price (-0.73); the **West** region is the most sensitive (-1.44). Pricing should not be set nationally as one number.

Customer Segmentation (K-Means)

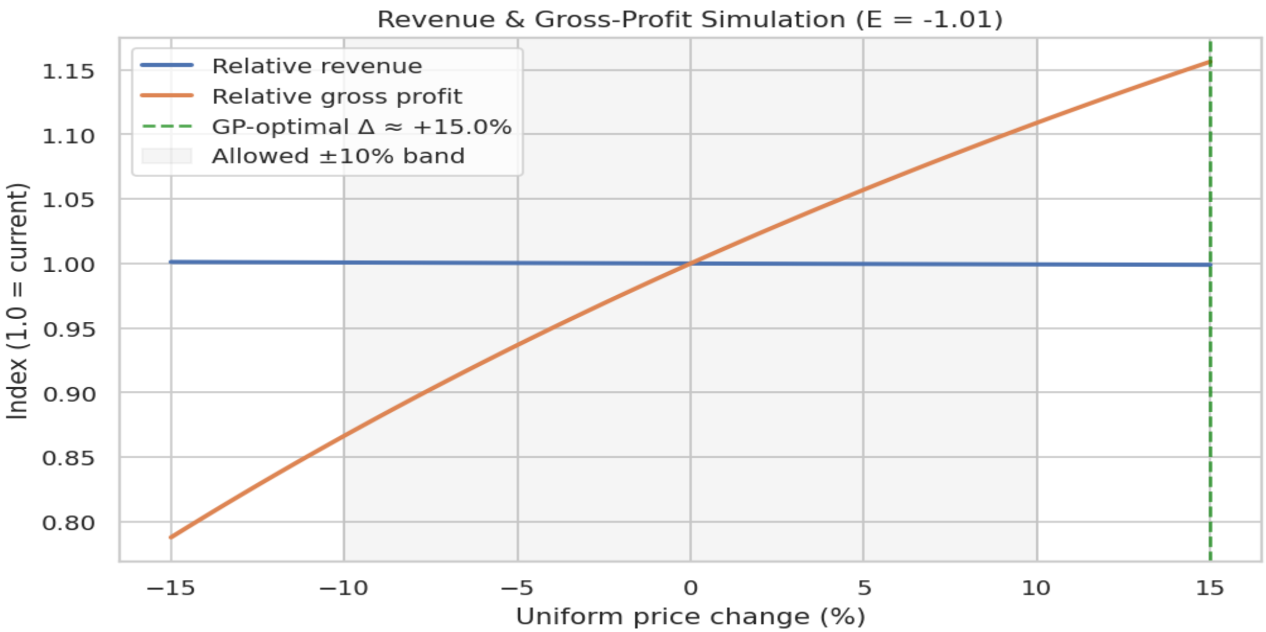


Customers cluster into distinct behavioural groups by discount appetite, price level and margin contribution — the basis for differentiated pricing and promotion rules rather than one-size-fits-all discounting.

Pricing Power by Customer Cluster

Elasticity re-estimated within each cluster confirms the segments are not just behaviourally different — they respond to price differently, which is exactly what a differentiated pricing strategy exploits.

Revenue & Gross-Profit Simulation

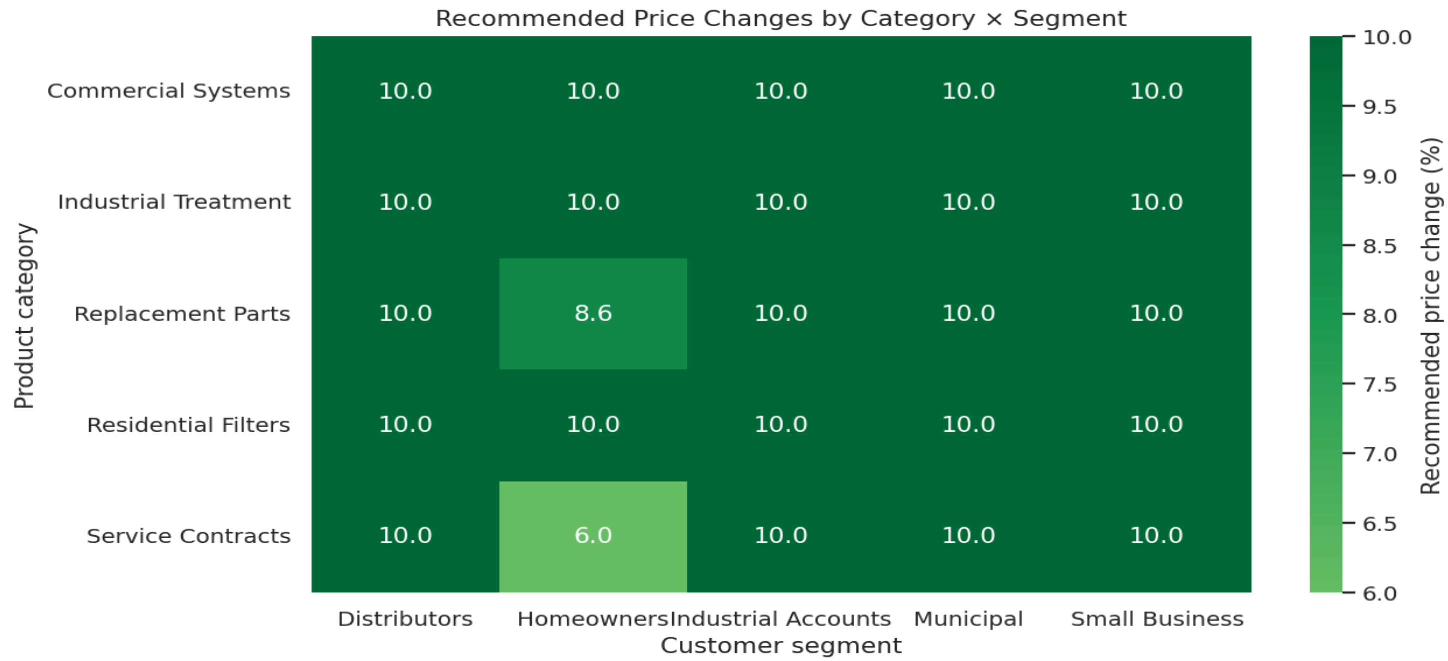


Before any change goes live, we simulate it. The gross-profit-maximising move differs from the revenue-maximising move — which is why optimizing the right objective matters. The shaded band marks the governance-approved ±10% range.

Optimization — Recommendation Mix

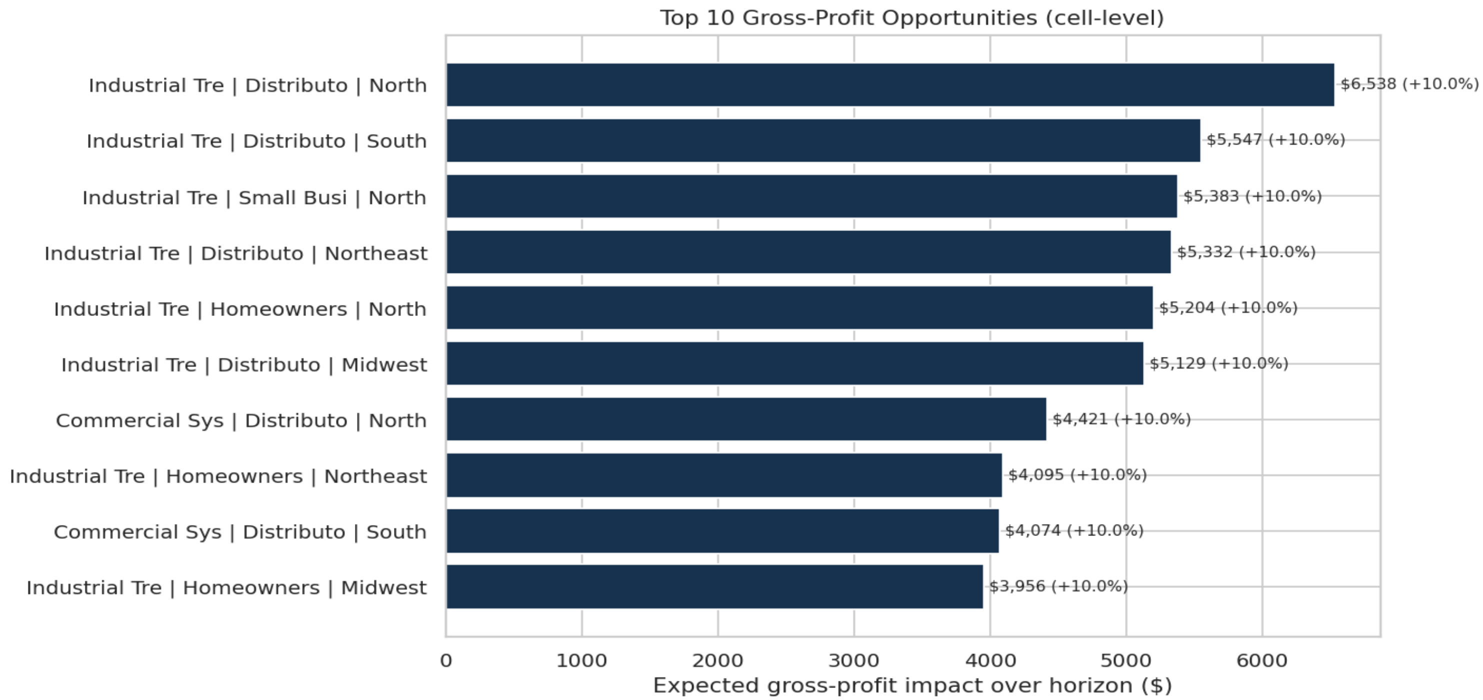
55 price-increase, 1 targeted-promotion, 0 discount-reduction, 69 hold and 0 manual-review actions — with their expected gross-profit contribution. Action is concentrated where elasticity is reliably low.

Optimization — Recommended Price Changes



Recommended net-price change by category × segment. Green = raise price (pricing power), red = reduce / promote (elastic). Every cell stays inside the ±10% governance band.

Where the Value Is — Top Opportunities



The ten highest-impact cells, with the recommended price move. These are the first actions to execute — highest gross-profit gain, lowest execution risk.

Business Impact & Client Value

- **Increase price where elasticity is low** — measured pricing power, not guesswork.
- **Reduce unnecessary discounts** where they fail to buy incremental volume.
- **Use targeted promotions only for elastic segments** that actually respond.
- **Monitor competitor-sensitive categories** via the competitor price index.
- **Run monthly price simulations** before approving any change.
- **Align Finance, Marketing & Sales** on one shared set of pricing rules.

Net effect: a defensible pricing system worth **$231,795** (**+10.1%** gross profit) while protecting volume in sensitive segments such as Homeowners.

Implementation Roadmap

Phase	Focus	Outcome
1 — Foundations	Connect sales, cost, competitor & macro data	Clean, governed pricing panel
2 — Measure	Re-estimate elasticity monthly	Live elasticity dashboard
3 — Optimize	Generate constrained price recommendations	Approved price bands by cell
4 — Govern	Finance / Marketing / Sales review cadence	Repeatable pricing governance

Start with the top-opportunity cells under a controlled pilot, measure realised vs. predicted impact, then scale across the portfolio.

Assumptions, Limitations & Data Note

- Demand follows a constant-elasticity (log-log) form over the analysed price range; extrapolation beyond ±10% is not advised.
- Elasticities are estimated from historical variation; structural market shifts require re-estimation.
- Low-confidence or unstable cells are flagged for manual review rather than auto-priced.
- Competitor effects are captured through category-level price indices, not every individual substitute.
- Dollar magnitudes scale with business size; figures here reflect the modelled company and horizon.

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